

Network Queueing Models Supplementary Notes

Richard T. B. Ma

School of Computing

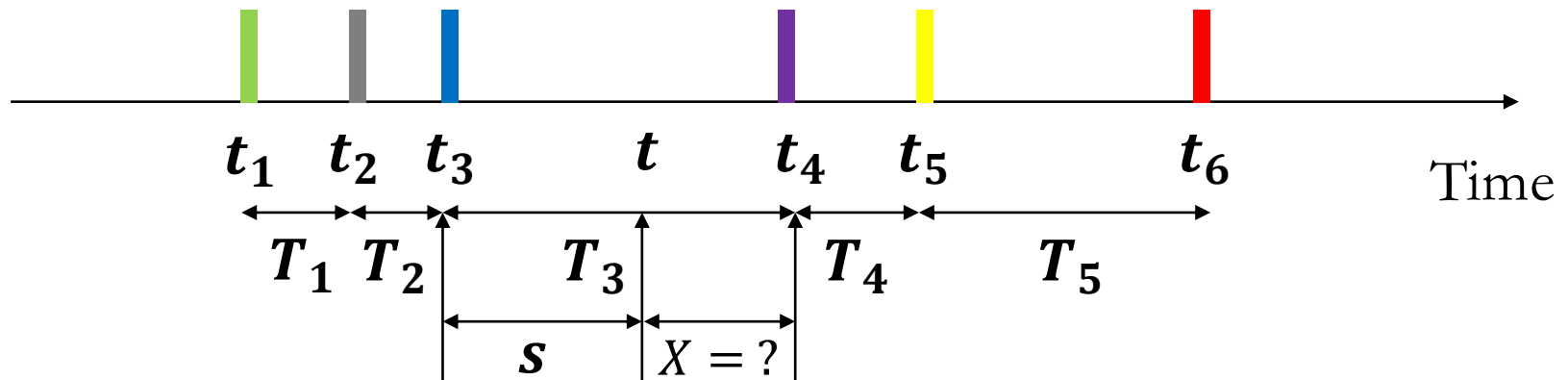
National University of Singapore

CS4226: Internet Architecture

Learning Outcomes

- ❑ What is a Poisson Process?
- ❑ What is the memoryless property?
- ❑ How to understand the merging property?
- ❑ What is an M/M/1 system and its state?
- ❑ What are the physical meaning of λ and μ ?
- ❑ What do we know about the utilization ρ ?
- ❑ Two definitions/meanings of π_i ?

Poisson and Memoryless Property



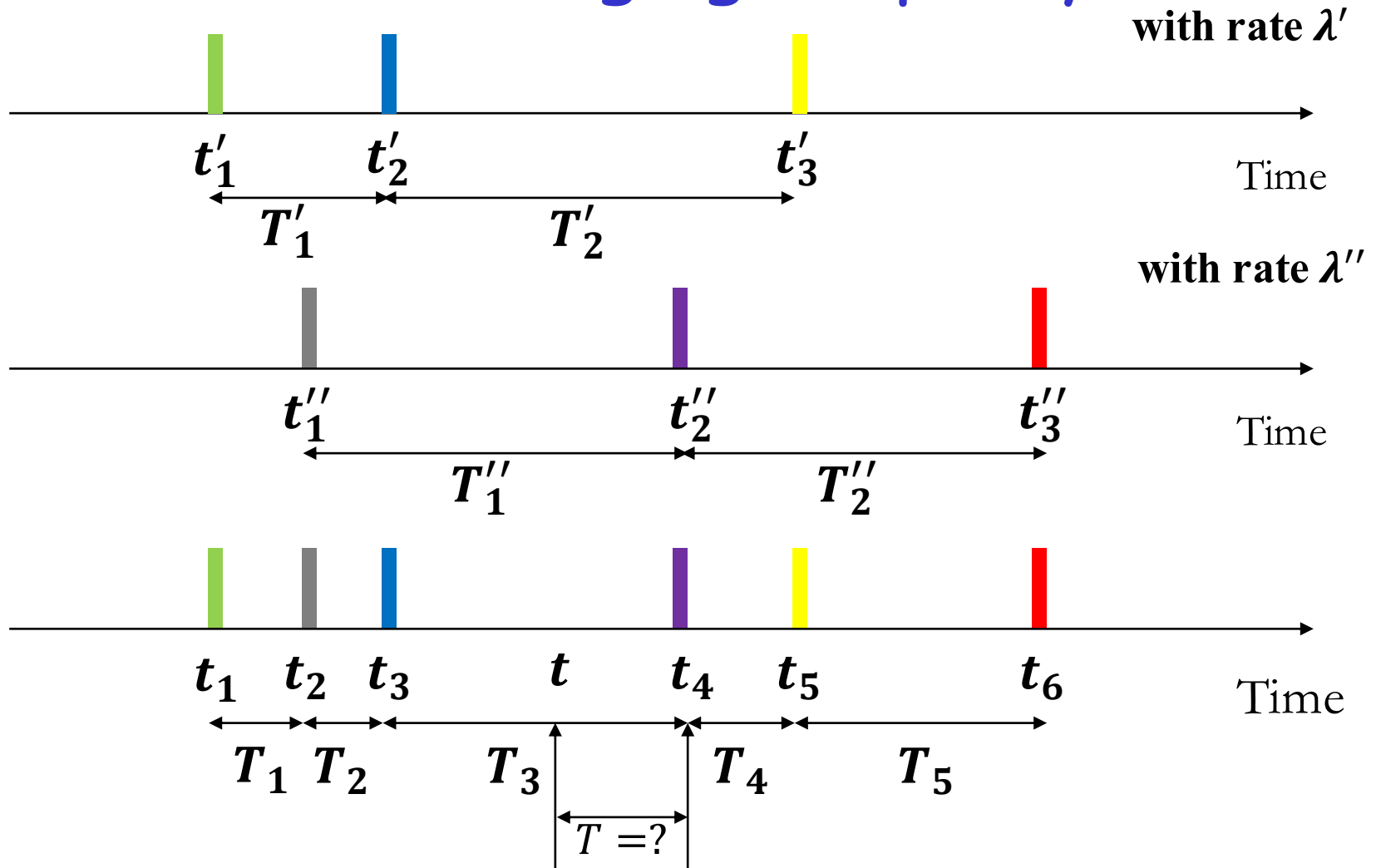
□ T_i are i.i.d. exponential with mean $1/\lambda$

□ What is the distribution of X ?

$$\diamond P\{T_3 - s > x \mid T_3 > s\} = P\{T_3 > s + x \mid T_3 > s\}$$

$$\diamond = P\{T_3 > x\}$$

Poisson and Merging Property

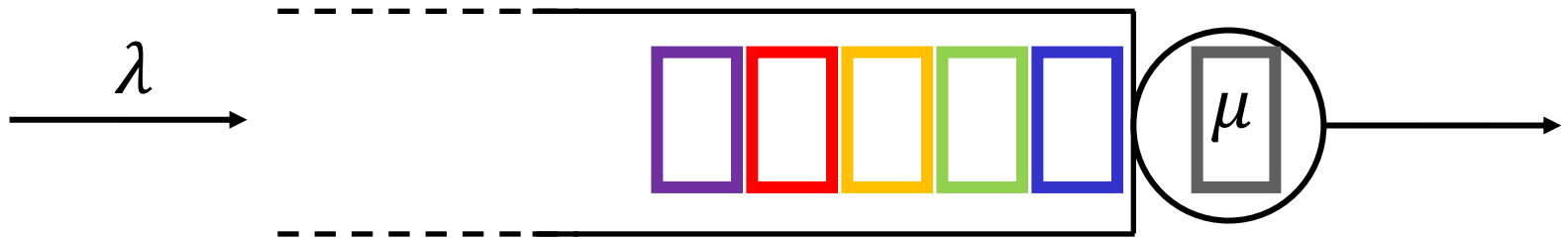


□ $T = \min\{T', T''\} \Rightarrow T$ is exp. with mean $\frac{1}{\lambda' + \lambda''}$

Learning Outcomes

- ❑ What is a Poisson Process?
- ❑ What is the memoryless property?
- ❑ How to understand the merging property?
- ❑ What is an M/M/1 system and its state?
- ❑ What are the physical meaning of λ and μ ?
- ❑ What do we know about the utilization ρ ?
- ❑ Two definitions/meanings of π_i ?

M/M/1 Model



- ❑ Single server with a queue of infinite size
- ❑ Poisson arrival with rate λ
- ❑ Exponential i.i.d. service time with rate μ
- ❑ Arrival and service times are independent
- ❑ First-In-First-Out (FIFO) service discipline

Is M/M/1 a good model to learn?

❑ Does real Internet traffic follow Poisson?

- ❖ Yes, by a classic paper "A Nonstationary Poisson View of Internet Traffic"
- ❖ No, it really depends on the application protocol

❑ Does service time follows exponential?

- ❖ Probably no, as TCP packets have a max size
- ❖ Maybe yes, for processing time of certain "services"

❑ Why is M/M/1 still good to learn?

- ❖ Simplest, your first step to more realistic models
- ❖ Provides macroscopic intuitions of system behaviors

Heavy Notation

□ What are they?

❖ t, t_i, t_i^d

❖ $W_i, T_i, W, D_i, E[D], S_i, E[S]$

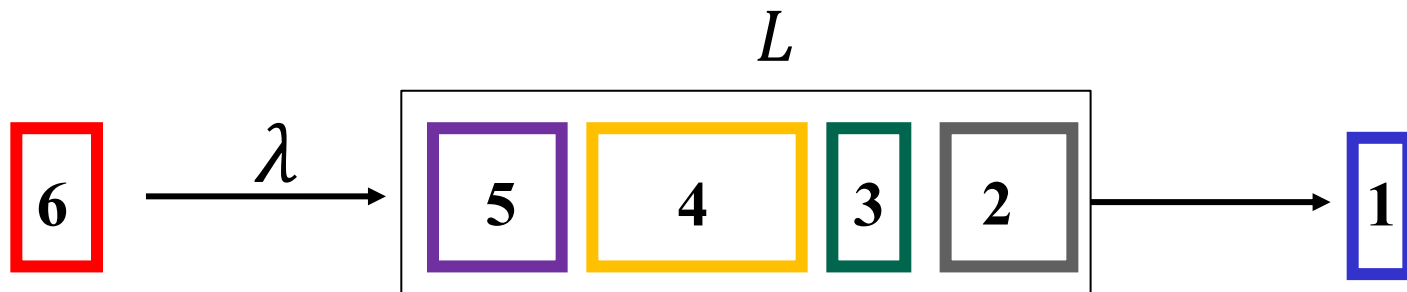
❖ $N(t), L(t), L, E[L], Q, E[Q]$

❖ λ, μ

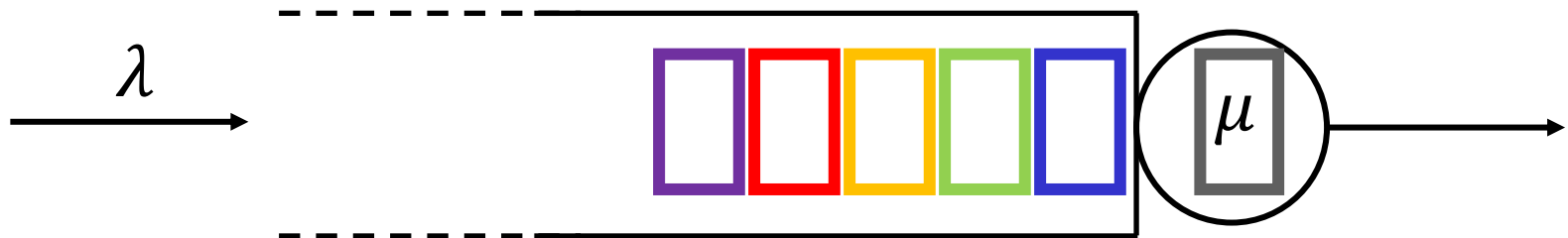
❖ ρ, π_i

Context: Two Related Systems

- A general system:



- A special system:



Heavy Notation

□ Times:

- ❖ time t and arrival time t_i
- ❖ inter-arrival time $T_i \stackrel{\text{def}}{=} t_{i+1} - t_i$
- ❖ sojourn time $W_i \stackrel{\text{def}}{=} t_i^d - t_i$,
- ❖ mean sojourn time
 - as a deterministic limit: $W \stackrel{\text{def}}{=} \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n W_j$
 - as a random variable: $E[W]$
- ❖ in a queueing system: $W_i = D_i + S_i$
 - mean queueing delay $E[D]$
 - mean service time $E[S]$

Heavy Notation

□ What are they?

❖ t, t_i, t_i^d

❖ $W_i, T_i, W, D_i, E[D], S_i, E[S]$

❖ $N(t), L(t), L, E[L], Q, E[Q]$

❖ λ, μ

❖ ρ, π_i

Heavy Notation

□ Numbers:

- ❖ # of packets/jobs/users $L(t)$
- ❖ average/mean # of packets in the system
 - as a deterministic limit: $L \stackrel{\text{def}}{=} \lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t L(s) ds$
 - as a random variable: $E[L]$
- ❖ in an M/M/1 queueing system: $E[L] = E[Q] + E[X]$
 - mean # of packets in the queue $E[Q]$
 - mean # of packets in the server $E[X]$

$$E[X] = P\{X = 0\} * 0 + P\{X = 1\} * 1 = P\{X = 1\}$$

$$P\{X = 1\} = P\{L - Q = 1\} = P\{L > 0\} = 1 - \pi_0 = \rho$$

Heavy Notation

□ What are they?

❖ t, t_i, t_i^d

❖ $W_i, T_i, W, D_i, E[D], S_i, E[S]$

❖ $N(t), L(t), L, E[L], Q, E[Q]$

❖ λ, μ

❖ ρ, π_i

Heavy Notation

□ Rates:

❖ arrival rate λ

- as a deterministic limit: $\lambda \stackrel{\text{def}}{=} \lim_{t \rightarrow \infty} \frac{N(t)}{t}$
- as a function of random variable: $\lambda \stackrel{\text{def}}{=} \frac{1}{E[T]}$

❖ service rate μ

- defined in a queueing system as: $\mu \stackrel{\text{def}}{=} \frac{1}{E[S]}$

□ Ratio, percentage, probability:

❖ in a single-server queueing system:

- % of time that the server is busy: ρ

❖ in M/M/1 system: $\pi = (\pi_0, \pi_1, \dots)$